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JOINT ENTRANCE EXAMINATION (MAIN) JULY -2022
(CANDIDATE RESPONSE SHEET)

Paper/Subject	B.E/B.Tech.(Paper 1)
Exam Date	28 Jul 2022
Exam Slot	1

Question ID:100601
Topic Name:Mathematics-Section A
Question:

Let the solution curve of the differential equation $x\mathrm{d}y=\left(\sqrt{x^2+y^2}+y\right)\mathrm{d}x$, $x>0$, intersect the line $x=1$ at $y=0$ and the line $x=2$ at $y=\alpha$. Then the value of α is :

A $\frac{1}{2}$

B $\frac{3}{2}$

C $-\frac{3}{2}$

D $\frac{5}{2}$

Answer Given By Candidate:Not

AttemptedQuestion ID:100602
Topic Name:Mathematics-Section A
Question:

Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x)=\cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is :

A $\left(-\infty,\frac{1}{4}\right]$

B $\left[-\frac{1}{4},\infty\right)$

C $\left(-1/3,\infty\right)$

D

$\left(-\infty, \frac{1}{3}\right]$

Answer Given By Candidate:Not Attempted

Question ID:100603
Topic Name:Mathematics-Section A
Question:

Let the vectors $\vec{a}=(1+t)\hat{i}+(1-t)\hat{j}+\hat{k}$, $\vec{b}=(1-t)\hat{i}+(1+t)\hat{j}+2\hat{k}$ and $\vec{c}=t\hat{i}-t\hat{j}+\hat{k}$, $t \in \mathbf{R}$ be such that for $\alpha, \beta, \gamma \in \mathbf{R}$, $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\vec{0} \Rightarrow \alpha=\beta=\gamma=0$. Then, the set of all values of t is :

A a non-empty finite set

B equal to $\mathbf{N}$

C equal to $\mathbf{R}-\{0\}$

D equal to $\mathbf{R}$

Answer Given By Candidate:D

Question ID:100604
Topic Name:Mathematics-Section A
Question:

Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x)-2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to :

A 0

B 1**C**

D $\frac{1}{-\frac{1}{2}}$

Answer Given By Candidate:Not

Attempted**Question ID:100605**
Topic Name:Mathematics-Section A
Question:

Let the operations $*, \odot \in \{\wedge, \vee\}$. If $(p*q)\odot(p \odot \sim q)$ is a tautology, then the ordered pair $(*, \odot)$ is :

A $(\vee, \wedge)$

B $(\vee, \vee)$

C $(\wedge, \wedge)$

D $(\wedge, \vee)$

Answer Given By Candidate:B

Question ID:100606
Topic Name:Mathematics-Section A
Question:

Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbf{R} \times \mathbf{R} - \{(0, 0)\}$, the vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$.

Then the value of $\left|\vec{a} \times \vec{b}\right|$ is equal to :

A $9\sqrt{3}$

B $27\sqrt{3}$

C 9

D 81

Answer Given By Candidate:B

Question ID:100607
Topic Name:Mathematics-Section A
Question:

For $t \in (0, 2\pi)$, if ABC is an equilateral triangle with vertices A(sint, $-$cost), B(cost, sint) and C(a, b) such that its orthocentre lies on a circle with centre $(1, \frac{1}{3})$, then $(a^2 - b^2)$ is equal to :

A $\frac{8}{3}$

B 8

C $\frac{77}{9}$

D $\frac{80}{9}$

Answer Given By Candidate:Not Attempted

Question ID:100608
Topic Name:Mathematics-Section A
Question:

For $\alpha \in \mathbf{N}$, consider a relation R on **N** given by $R = \{(x, y) : 3x + \alpha y \text{ is a multiple of } 7\}$. The relation R is an equivalence relation if and only if :

A $\alpha = 14$

B α is a multiple of 4

C 4 is the remainder when α is divided by 10

D 4 is the remainder when α is divided by 7

Answer Given By Candidate:D

Question ID:100609
Topic Name:Mathematics-Section A
Question:

Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is :

A $\frac{3}{4}$

B $\frac{11}{16}$

C $\frac{23}{32}$

D $\frac{13}{16}$

Answer Given By Candidate:A

Question ID:100610
Topic Name:Mathematics-Section A
Question:

If $y = y(x)$, $x \in (0, \pi/2)$ be the solution curve of the differential equation $\left(\sin^2 2x\right)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y = 2e^{-4x}(2\sin 2x + \cos 2x)$, with $y(\pi/4) = e^{-\pi}$, then $y(\pi/6)$ is equal to :

A

$\frac{2}{\sqrt{3}}e^{-2\pi/3}$

B $\frac{2}{\sqrt{3}}e^{2\pi/3}$

C $\frac{1}{\sqrt{3}}e^{-2\pi/3}$

D $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Answer Given By Candidate: Not Attempted

Question ID: 100611
Topic Name: Mathematics-Section A
Question:

If the tangents drawn at the points P and Q on the parabola $y^2 = 2x-3$ intersect at the point R(0, 1), then the orthocentre of the triangle PQR is :

A (0, 1)

B (2, −1)

C (6, 3)

D (2, 1)

Answer Given By Candidate: Not Attempted

Question ID: 100612
Topic Name: Mathematics-Section A
Question:

Let C be the centre of the circle $x^2+y^2-x+2y = \frac{11}{4}$ and P be a point on the circle. A line passes through the point C, makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at the points Q and R. Then the area of the triangle PQR (in unit^2) is :

A 2

B $2\sqrt{2}$

C $8\sin\left(\frac{\pi}{8}\right)$

D

$8\cos\left(\frac{\pi}{8}\right)$

Answer Given By Candidate: Not Attempted

Question ID: 100613
Topic Name: Mathematics-Section A
Question: The remainder when $7^{2022}+3^{2022}$ is divided by 5 is :

A 0

B 2

C 3

D 4

Answer Given By Candidate: B

Question ID: 100614
Topic Name: Mathematics-Section A
Question:

Let the matrix $A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ and the matrix $B_0=A^{49}+2A^{98}$. If $B_n=\text{Adj}(B_{n-1})$ for all $n \geq 1$, then $\det(B_4)$ is equal to :

A 3^{28}

B 3^{30}

C 3^{32}

D 3^{36}

Answer Given By Candidate: C

Question ID: 100615
Topic Name: Mathematics-Section A
Question:

Let $S_1=\left\{z_1 \in \mathbf{C}: |z_1-3|=\frac{1}{2}\right\}$ and $S_2=\left\{z_2 \in \mathbf{C}: |z_2-|z_2+1||=|z_2+|z_2-1|\right\}$. Then, for $z_1 \in S_1$ and $z_2 \in S_2$, the least value of $|z_2-z_1|$ is :

A 0

B $\frac{1}{2}$

C $\frac{3}{2}$

D $\frac{5}{2}$

Answer Given By Candidate:Not Attempted

Question ID:100616
Topic Name:Mathematics-Section A
Question:

The foot of the perpendicular from a point on the circle $x^2+y^2=1$, z = 0 to the plane $2x+3y+z = 6$ lies on which one of the following curves ?

A $(6x+5y-12)^2+4(3x+7y-8)^2=1,\ z=6-2x-3y$

B $(5x+6y-12)^2+4(3x+5y-9)^2=1,\ z=6-2x-3y$

C $(6x+5y-14)^2+9(3x+5y-7)^2=1,\ z=6-2x-3y$

D $(5x+6y-14)^2+9(3x+7y-8)^2=1,\ z=6-2x-3y$

Answer Given By Candidate:Not Attempted

Question ID:100617
Topic Name:Mathematics-Section A
Question:

If the minimum value of $f(x)=\frac{5x^2}{2}+\frac{\alpha}{x^5}$, $x > 0$, is 14, then the value of α is equal to :

A 32

B 64

C 128

D 256

Answer Given By Candidate:C

Question ID:100618
Topic Name:Mathematics-Section A
Question:

Let α, β and γ be three positive real numbers. Let $f(x) = \alpha x^5+\beta x^3+\gamma x$, $x \in \mathbf{R}$ and $g : \mathbf{R}\to\mathbf{R}$ be such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. If $a_1, a_2, a_3, \ldots, a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$ is equal to :

A 0

B 3

C 9

D 27

Answer Given By Candidate:A

Question ID:100619
Topic Name:Mathematics-Section A
Question:

Consider the sequence $a_1, a_2, a_3, \ldots$ such that $a_1 = 1$, $a_2 = 2$ and $a_{n+2} = \frac{2}{a_{n+1}} + a_n$ for $n = 1, 2, 3, \ldots$. If $\left(\frac{a_1 + \frac{1}{a_2}}{a_3}\right)\cdot\left(\frac{a_2 + \frac{1}{a_3}}{a_4}\right)\cdot\left(\frac{a_3 + \frac{1}{a_4}}{a_5}\right)\ldots\left(\frac{a_{30} + \frac{1}{a_{31}}}{a_{32}}\right) = 2^{\alpha}\left({}^{61}C_{31}\right)$, then α is equal to :

A -30

B -31

C -60

D -61

Answer Given By Candidate:Not Attempted

Question ID:100620
Topic Name:Mathematics-Section A
Question:

The minimum value of the twice differentiable function $f(x) = \int_0^x e^{x-t} f'(t)dt - (x^2 - x + 1)e^x$, $x \in \mathbf{R}$, is :

A $-\frac{2}{\sqrt{e}}$

B $-2\sqrt{e}$

C $-\sqrt{e}$

D $\frac{2}{\sqrt{e}}$

Answer Given By Candidate:Not Attempted

Question ID:100621
Topic Name:Mathematics-Section B
Question:

Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from {A, B, C, D, E} or a number from {1, 2, 3, 4, 5} with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from {1, 2, 3, 4, 5} is $\alpha \times 5^6$, then α is equal to _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100622
Topic Name:Mathematics-Section B
Question:

Let $P(-2, -1, 1)$ and $Q\left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagonal RS are α, -1, β, where both α and β are integers of minimum absolute values, then $\alpha^2+\beta^2$ is equal to _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100623
Topic Name:Mathematics-Section B
Question:

Let $f: [0, 1] \to \mathbf{R}$ be a twice differentiable function in (0, 1) such that $f(0) = 3$ and $f(1) = 5$. If the line $y = 2x + 3$ intersects the graph of f at only two distinct points in (0, 1), then the least number of points $x \in (0, 1)$, at which $f''(x) = 0$, is _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100624
Topic Name:Mathematics-Section B
Question:

If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}dx = \alpha\sqrt{2}+\beta\sqrt{3}$, where α, β are integers, then $\alpha+\beta$ is equal to _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100625
Topic Name:Mathematics-Section B

Question:

Let $A=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}$, $\alpha, \beta \in \mathbf{R}$. Let α_1 be the value of α which satisfies $(A+B)^2=A^2+\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$ and α_2 be the value of α which satisfies $(A+B)^2 = B^2$. Then $|\alpha_1-\alpha_2|$ is equal to _____.

Answer Given By Candidate:10

Question ID:100626
Topic Name:Mathematics-Section B
Question:

For $p, q \in \mathbf{R}$, consider the real valued function $f(x) = (x-p)^2-q$, $x \in \mathbf{R}$ and $q > 0$. Let a_1, a_2, a_3 and a_4 be in an arithmetic progression with mean p and positive common difference. If $|f(a_i)|=500$ for all $i = 1, 2, 3, 4$, then the absolute difference between the roots of $f(x) = 0$ is _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100627
Topic Name:Mathematics-Section B
Question:

For the hyperbola $H : x^2-y^2 = 1$ and the ellipse $E : \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a > b > 0$, let the

(1) eccentricity of E be reciprocal of the eccentricity of H, and

(2) the line $y=\sqrt{\frac{5}{2}}x+K$ be a common tangent of E and H.

Then $4(a^2+b^2)$ is equal to _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100628
Topic Name:Mathematics-Section B
Question:

Let $x_1, x_2, x_3, ..., x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i-i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is _____.

Answer Given By Candidate:10

Question ID:100629
Topic Name:Mathematics-Section B

Question:

$$\lim_{x\to 0}\left(\frac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$$ is equal to _____.

Answer Given By Candidate: Not Attempted

Question ID: 100630
Topic Name: Mathematics-Section B
Question:

The sum of all real values of x for which $\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$ is equal to _____.

Answer Given By Candidate: Not

Attempted **Question ID:** 100631
Topic Name: Physics-Section A

Question: The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be : (if μ_0 : permeability of free space and B : magnetic field)

A $[M\,L^2\,T^{-2}]$

B $[M\,L\,T^{-2}]$

C $[M\,L^{-1}\,T^{-2}]$

D $[M\,L^2\,T^{-2}\,A^{-1}]$

Answer Given By Candidate: C

Question ID: 100632
Topic Name: Physics-Section A
Question:

A NCC parade is going at a uniform speed of 9 km/h under a mango tree on which a monkey is sitting at a height of 19.6 m. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is : (Given $g=9.8\ m/s^2$)

A 5 m

B 10 m

C 19.8 m

D 24.5 m

Answer Given By Candidate:A

Question ID:100633
Topic Name:Physics-Section A
Question:

In two different experiments, an object of mass 5 kg moving with a speed of 25 ms^{-1} hits two different walls and comes to rest within (i) 3 second, (ii) 5 seconds, respectively. Choose the **correct** option out of the following :

A Impulse and average force acting on the object will be same for both the cases.

B Impulse will be same for both the cases but the average force will be different.

C Average force will be same for both the cases but the impulse will be different.

D Average force and impulse will be different for both the cases.

Answer Given By Candidate:B

Question ID:100634
Topic Name:Physics-Section A
Question:

A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm/s. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne).

A 3

B 9

C 12

D 18

Answer Given By Candidate:Not Attempted

Question ID:100635
Topic Name:Physics-Section A
Question:

If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately :

A decrease by 2%

B decrease by 4%

C increase by 2%

D increase by 4%

Answer Given By Candidate:D

Question ID:100636
Topic Name:Physics-Section A
Question:

The force required to stretch a wire of cross-section 1 cm^2 to double its length will be : (Given Yong's modulus of the wire $=2\times10^{11}\ N/m^2$)

A $1\times10^7\ N$

B $1.5\times10^7\ N$

C $2\times10^7\ N$

D $2.5\times10^7\ N$

Answer Given By Candidate:C

Question ID:100637
Topic Name:Physics-Section A
Question:

A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by 40°C, its efficiency increases by 30%. The temperature of the source will be :

A 166.7 K

B 255.1 K

C 266.7 K

D 367.7 K

Answer Given By Candidate:C

Question ID:100638
Topic Name:Physics-Section A
Question:

Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become $2v$.

In the light of the above statements, choose the **correct** answer from the options given below :

A Both **Statement I** and **Statement II** are true

B Both **Statement I** and **Statement II** are false

C **Statement I** is true but **Statement II** is false

D **Statement I** is false but **Statement II** is true

Answer Given By Candidate:A

Question ID:100639
Topic Name:Physics-Section A
Question:

In the wave equation

$$y = 0.5 \sin \frac{2\pi}{\lambda} (400\ t - x)\ m$$

the velocity of the wave will be :

A 200 m/s

B $200\sqrt{2}$ m/s

C 400 m/s

D $400\sqrt{2}$ m/s

Answer Given By Candidate:A

Question ID:100640
Topic Name:Physics-Section A
Question:

Two capacitors, each having capacitance 40 μF are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became 24 μF. The value of K will be :

A 1.5

B 2.5

C 1.2

D 3

Answer Given By Candidate:C

Question ID:100641
Topic Name:Physics-Section A
Question:

A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is :

A 9 : 1

B 1 : 9

C 4 : 1

D

3 : 1

Answer Given By Candidate:C

Question ID:100642
Topic Name:Physics-Section A

The current sensitivity of a galvanometer can be increased by :

(A) decreasing the number of turns

(B) increasing the magnetic field

(C) decreasing the area of the coil

(D) decreasing the torsional constant of the spring

Question: Choose the **most appropriate** answer from the options given below :

A (B) and (C) only

B (C) and (D) only

C (A) and (C) only

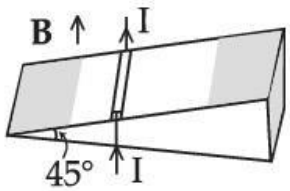
D (B) and (D) only

Answer Given By Candidate:D

Question ID:100643
Topic Name:Physics-Section A
Question:

As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be :

{Use $g = 10\ m/s^2$}



A 30 A

B 15 A

C 10 A

D 3 A

Answer Given By Candidate:A

Question ID:100644
Topic Name:Physics-Section A

Question:

The equation of current in a purely inductive circuit is $5\sin(49\ \pi t - 30°)$. If the inductance is 30 mH then the equation for the voltage across the inductor, will be :

$\left\{\text{Let } \pi = \frac{22}{7}\right\}$

A $1.47\ \sin(49\ \pi t - 30°)$

B $1.47\ \sin(49\ \pi t + 60°)$

C $23.1\ \sin(49\ \pi t - 30°)$

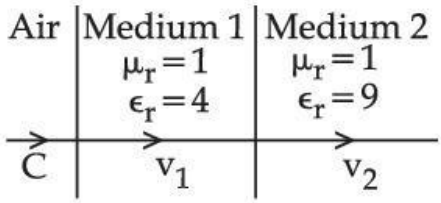
D $23.1\ \sin(49\ \pi t + 60°)$

Answer Given By Candidate:D

Question ID:100645
Topic Name:Physics-Section A
Question:

As shown in the figure, after passing through the medium 1. The speed of light v_2 in medium 2 will be :

(Given $c = 3 \times 10^8\ ms^{-1}$)



A $1.0 \times 10^8\ ms^{-1}$

B $0.5 \times 10^8\ ms^{-1}$

C $1.5 \times 10^8\ ms^{-1}$

D $3.0 \times 10^8\ ms^{-1}$

Answer Given By Candidate:B

Question ID:100646
Topic Name:Physics-Section A
Question:

In normal adujstment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2, will be :

A 20 cm

B 30 cm

C

10 cm

D 15 cm

Answer Given By Candidate:D

Question ID:100647
Topic Name:Physics-Section A
Question:

The equation $\lambda = \frac{1.227}{x}$ nm can be used to find the de-Brogli wavelength of an electron.

In this equation x stands for :

Where m = mass of electron
P = momentum of electron
K = Kinetic energy of electron
V = Accelerating potential in volts for electron

A $\sqrt{mK}$

B $\sqrt{P}$

C $\sqrt{K}$

D $\sqrt{V}$

Answer Given By Candidate:A

Question ID:100648
Topic Name:Physics-Section A
Question:

The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$th of its original mass to disintegrate will be :

A 120 days

B 130 days

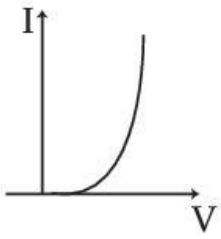
C 180 days

D 20 days

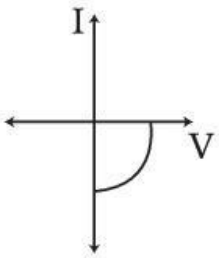
Answer Given By Candidate:C

Question ID:100649
Topic Name:Physics-Section A
Question: Identify the solar cell characteristics from the following options :

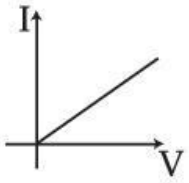
A



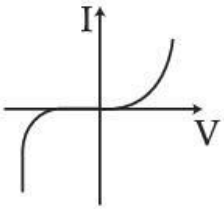
B



C



D



Answer Given By Candidate:D

Question ID:100650
Topic Name:Physics-Section A
Question:

In the case of amplitude modulation to avoid distortion the modulation index (μ) should be :

A $\mu \leq 1$

B $\mu \geq 1$

C $\mu = 2$

D $\mu = 0$

Answer Given By Candidate:A

Question ID:100651
Topic Name:Physics-Section B
Question:

If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be _____.

Answer Given By Candidate:5

Question ID:100652
Topic Name:Physics-Section B
Question:

A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours.

Answer Given By Candidate:15

Question ID:100653
Topic Name:Physics-Section B
Question:

In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm.

Answer Given By

Candidate:630Question

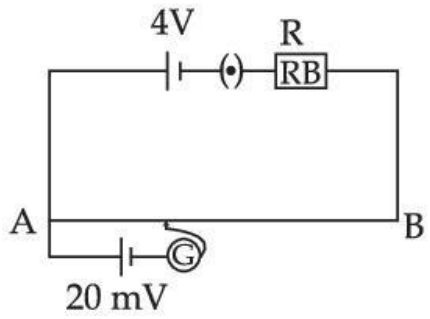
ID:100654
Topic Name:Physics-Section B
Question:

The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rad s^{-1} and 232 rad s^{-1}. The value of resistance in the circuit is $R = 5\ \Omega$. The self inductance in the circuit is _____ mH.

Answer Given By Candidate:Not

AttemptedQuestion ID:100655
Topic Name:Physics-Section B
Question:

As shown in the figure, a potentiometer wire of resistance 20 Ω and length 300 cm is connected with resistance box (R.B.) and a standard cell of emf 4 V. For a resistance 'R' of resistance box introduced into the circuit, the null point for a cell of 20 mV is found to be 60 cm. The value of 'R' is _____ Ω.



Answer Given By Candidate:780

Question ID:100656
Topic Name:Physics-Section B
Question:

Two electric dipoles of dipole moments 1.2×10^{-30} Cm and 2.4×10^{-30} Cm are placed in two different uniform electric fields of strengths 5×10^{4} NC^{-1} and 15×10^{4} NC^{-1} respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is _____.

Answer Given By Candidate:6

Question ID:100657
Topic Name:Physics-Section B
Question:

The frequency of echo will be _____ Hz if the train blowing a whistle of frequency 320 Hz is moving with a velocity of 36 km/h towards a hill from which an echo is heard by the train driver. Velocity of sound in air is 330 m/s.

Answer Given By Candidate:Not

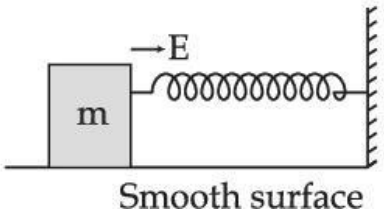
AttemptedQuestion ID:100658
Topic Name:Physics-Section B
Question:

The diameter of an air bubble which was initially 2 mm, rises steadily through a solution of density 1750 kg m^{-3} at the rate of 0.35 cms^{-1}. The coefficient of viscosity of the solution is _____ poise (in nearest integer). (the density of air is negligible).

Answer Given By Candidate:Not

AttemptedQuestion ID:100659
Topic Name:Physics-Section B
Question:

A block of mass 'm' (as shown in figure) moving with kinetic energy E compresses a spring through a distance 25 cm when, its speed is halved. The value of spring constant of used spring will be nE Nm^{-1} for n = _____.

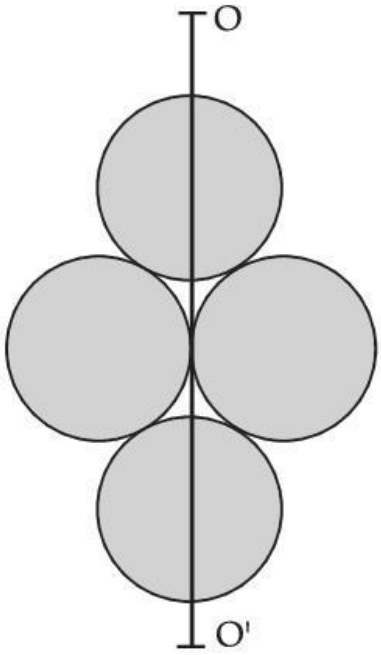


Answer Given By Candidate:Not

AttemptedQuestion ID:100660
Topic Name:Physics-Section B

Question:

Four identical discs each of mass 'M' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}Ma^2$. Then, the value of x will be _____.



Answer Given By Candidate:Not

AttemptedQuestion ID:100661
Topic Name:Chemistry-Section A

Question: Identify the incorrect statement from the following.

A A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.

B An orbital is the one electron wave function (ψ) in an atom.

C The existence of Bohr's orbits is supported by hydrogen spectrum.

D Atomic orbital is characterised by the quantum numbers n and l only.

Answer Given By Candidate:B

Question ID:100662
Topic Name:Chemistry-Section A

Question: Which of the following relation is not correct ?

A $\triangle H = \triangle U - P\triangle V$

B $\triangle U = q + W$

C

$\triangle S_{sys} + \triangle S_{surr} \geq 0$

D $\triangle G = \triangle H - T\triangle S$

Answer Given By Candidate:C

Question ID:100663
Topic Name:Chemistry-Section A
Question:

Match **List - I** with **List - II**.

	List - I		**List - II**
(A)	$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$	(I)	Primary battery
(B)	$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$	(II)	Discharging of secondary battery
(C)	$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$	(III)	Fuel cell
(D)	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	(IV)	Charging of secondary battery

Choose the **correct** answer from the options given below :

A (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

B (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

C (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

D (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Answer Given By Candidate:C

Question ID:100664
Topic Name:Chemistry-Section A

Match **List - I** with **List - II**.

	List - I **Reaction**		**List - II** **Catalyst**
(A)	$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$	(I)	$NO(g)$
(B)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(II)	$H_2SO_4(l)$
(C)	$C_{12}H_{22}O_{11}(aq) + H_2O(l) \rightarrow \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$	(III)	Pt(s)
(D)	$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(IV)	Fe(s)

Question: Choose the **correct** answer from the options given below :

A (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

B (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

C (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

D (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Answer Given By Candidate:C

Question ID:100665
Topic Name:Chemistry-Section A
Question:

In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical ?

(A) Rb and Cs

(B) Na and K

(C) Ar and Kr

(D) I and At

Choose the **correct** answer from the options given below :

A (A) and (B) only

B (B) and (C) only

C (A) and (C) only

D (C) and (D) only

Answer Given By Candidate:C

Question ID:100666
Topic Name:Chemistry-Section A
Question: Which of the reaction is suitable for concentrating ore by leaching process ?

A $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

B $Fe_3O_4 + CO \rightarrow 3FeO + CO_2$

C $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

D $Al_2O_3 + 6Mg \rightarrow 6MgO + 4Al$

Answer Given By Candidate:C

Question ID:100667
Topic Name:Chemistry-Section A

Question:

The metal salts formed during softening of hardwater using Clark's method are :

A $Ca(OH)_2$ and $Mg(OH)_2$

B $CaCO_3$ and $Mg(OH)_2$

C $Ca(OH)_2$ and $MgCO_3$

D $CaCO_3$ and $MgCO_3$

Answer Given By Candidate:B

Question ID:100668
Topic Name:Chemistry-Section A

Question: Which of the following statement is incorrect ?

A Low solubility of LiF in water is due to its small hydration enthalpy.

B KO_2 is paramagnetic.

C Solution of sodium in liquid ammonia is conducting in nature.

D Sodium metal has higher density than potassium metal.

Answer Given By Candidate:C

Question ID:100669
Topic Name:Chemistry-Section A

Match **List - I** with **List - II**, match the gas evolved during each reaction.

	List - I		**List - II**
(A)	$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(I)	H_2
(B)	$KMnO_4 + HCl \rightarrow$	(II)	N_2
(C)	$Al + NaOH + H_2O \rightarrow$	(III)	O_2
(D)	$NaNO_3 \xrightarrow{\Delta}$	(IV)	Cl_2

Question: Choose the **correct** answer from the options given below :

A (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

B (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

C (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

D

(A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Answer Given By Candidate:C

Question ID:100670
Topic Name:Chemistry-Section A
Question: Which of the following has least tendency to liberate H_2 from mineral acids ?

A Cu

B Mn

C Ni

D Zn

Answer Given By Candidate:A

Question ID:100671
Topic Name:Chemistry-Section A
Question:

Given below are two statements :

Statement I : In polluted water values of both dissolved oxygen and BOD are very low.

Statement II : Eutrophication results in decrease in the amount of dissolved oxygen.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

A Both **Statement I** and **Statement II** are true

B Both **Statement I** and **Statement II** are false

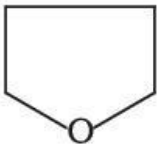
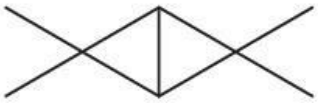
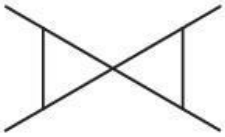
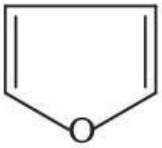
C **Statement I** is true but **Statement II** is false

D **Statement I** is false but **Statement II** is true

Answer Given By Candidate:D

Question ID:100672
Topic Name:Chemistry-Section A

Match **List - I** with **List - II**.

List - I		List - II
(A)		(I) Spiro compound
(B)		(II) Aromatic compound
(C)		(III) Non-planar Heterocyclic compound
(D)		(IV) Bicyclo compound

Question: Choose the **correct** answer from the options given below :

A (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

B (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

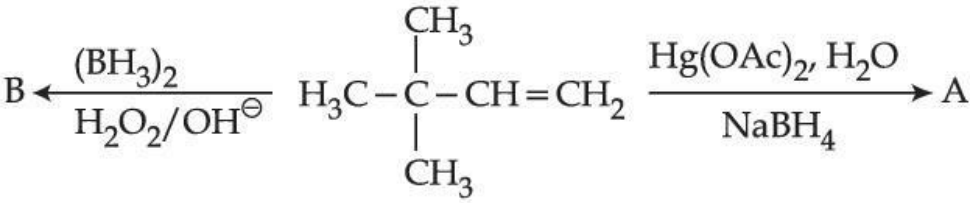
C (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

D (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Answer Given By Candidate:C

Question ID:100673
Topic Name:Chemistry-Section A

Choose the correct option for the following reactions.



Question:

A 'A' and 'B' are both Markovnikov addition products.

B 'A' is Markovnikov product and 'B' is anti-Markovnikov product.

C 'A' and 'B' are both anti-Markovnikov products.

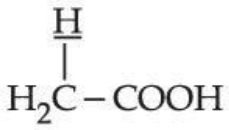
D 'B' is Markovnikov and 'A' is anti-Markovnikov product.

Answer Given By Candidate:B

Question ID:100674
Topic Name:Chemistry-Section A
Question:

Among the following marked proton of which compound shows lowest pK_a value ?

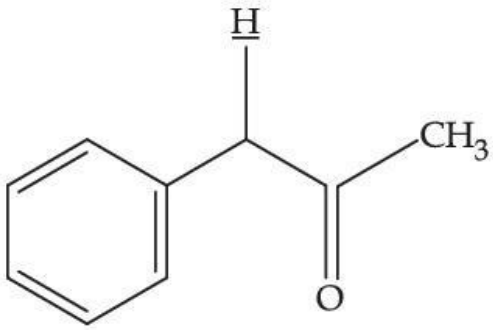
A



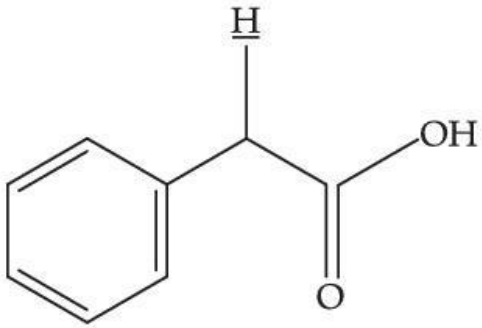
B



C



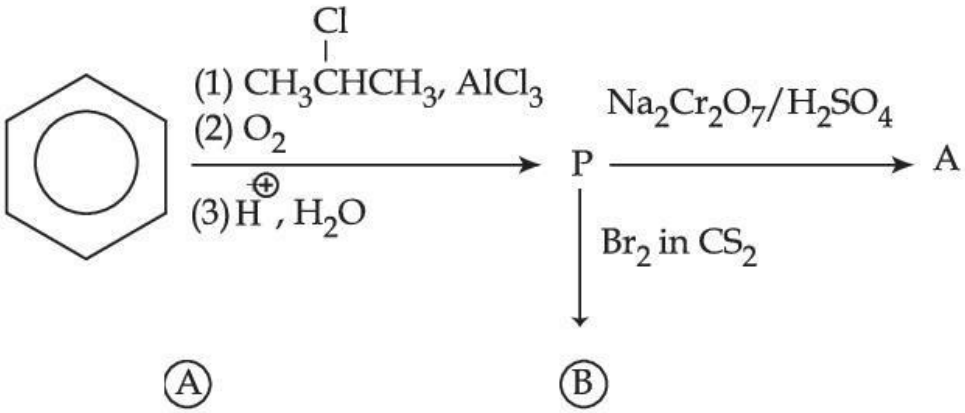
D



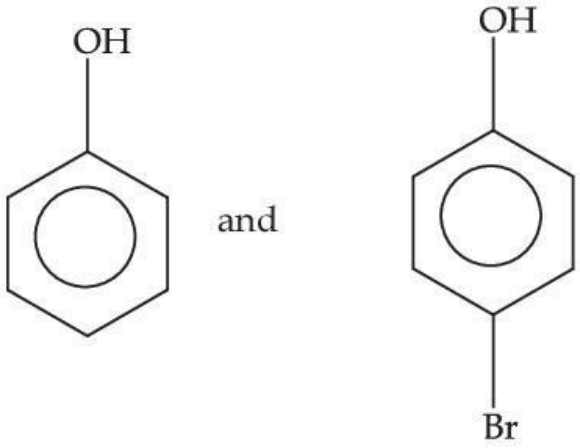
Answer Given By Candidate:C

Question ID:100675
Topic Name:Chemistry-Section A

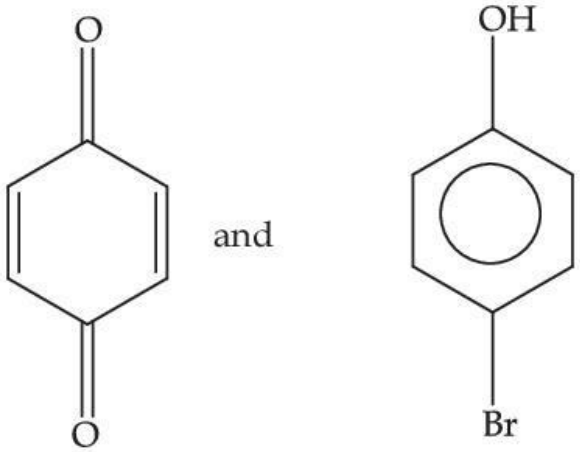
Identify the major products A and B for the below given reaction sequence.



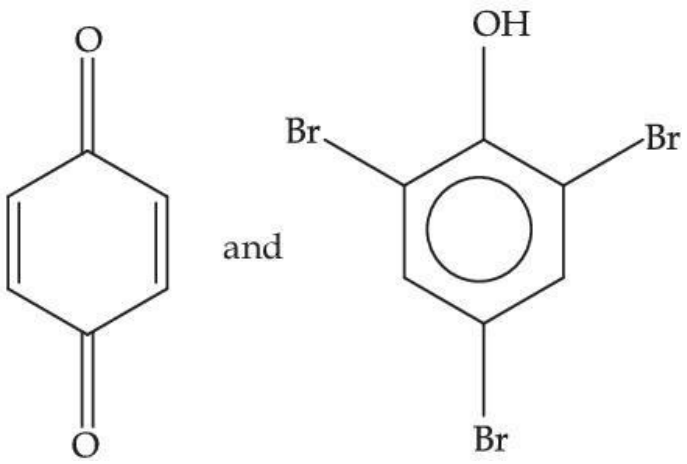
Question:
A



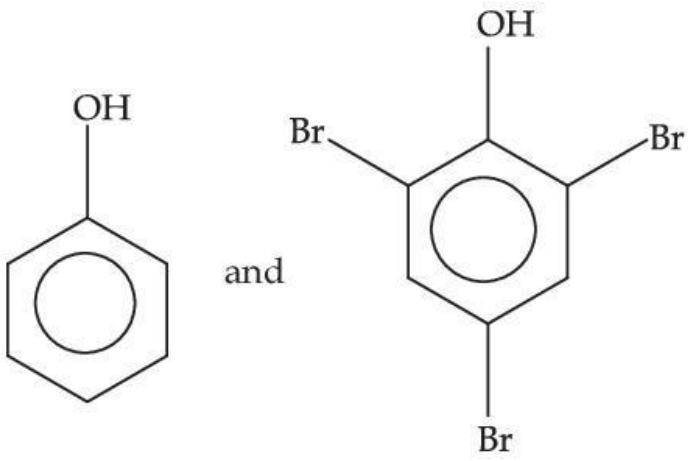
B



C



D

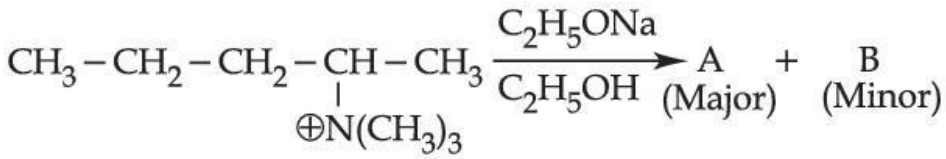


Answer Given By Candidate:B

Question ID:100676
Topic Name:Chemistry-Section A

Question:

Identify the correct statement for the below given transformation.



A A - $CH_3CH_2CH=CH-CH_3$, B - $CH_3CH_2CH_2CH=CH_2$, Saytzeff products

B A - $CH_3CH_2CH=CH-CH_3$, B - $CH_3CH_2CH_2CH=CH_2$, Hofmann products

C A - $CH_3CH_2CH_2CH=CH_2$, B - $CH_3CH_2CH=CHCH_3$, Hofmann products

D A - $CH_3CH_2CH_2CH=CH_2$, B - $CH_3CH_2CH=CHCH_3$, Saytzeff products

Answer Given By Candidate:A

Question ID:100677
Topic Name:Chemistry-Section A

Question: Terylene polymer is obtained by condensation of :

A Ethane-1, 2-diol and Benzene-1, 3 dicarboxylic acid

B Propane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

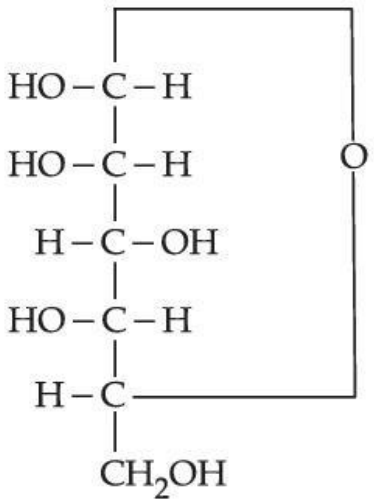
C Ethane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

D Ethane-1, 2-diol and Benzene-1, 2 dicarboxylic acid

Answer Given By Candidate:C

Question ID:100678
Topic Name:Chemistry-Section A

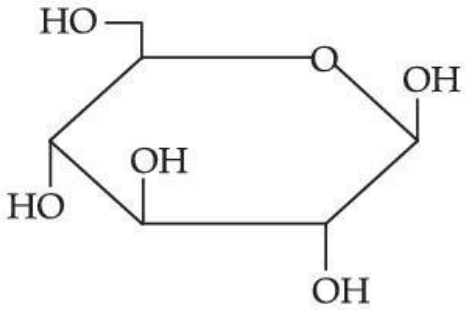
For the below given cyclic hemiacetal (X), the correct pyranose structure is :



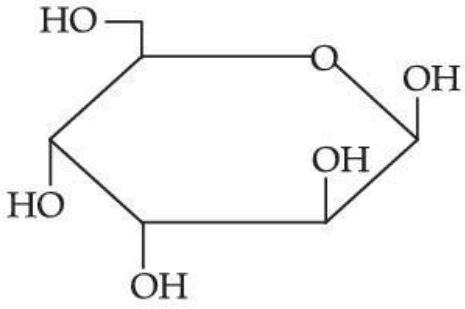
(X)

Question:

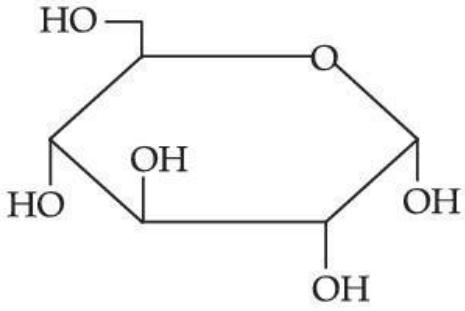
A



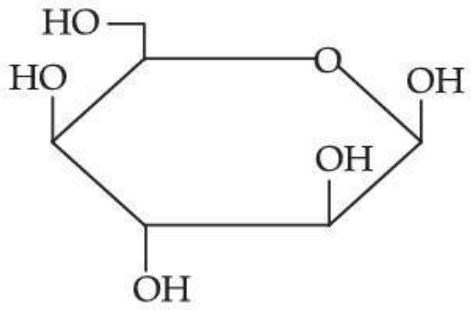
B



C



D



Answer Given By Candidate:C

Question ID:100679
Topic Name:Chemistry-Section A

Statements about Enzyme Inhibitor Drugs are given below :

(A) There are Competitive and Non-competitive inhibitor drugs.

(B) These can bind at the active sites and allosteric sites.

(C) Competitive Drugs are allosteric site blocking drugs.

(D) Non-competitive Drugs are active site blocking drugs.

Question: Choose the **correct** answer from the options given below :

A (A), (D) only

B (A), (C) only

C (A), (B) only

D (A), (B), (C) only

Answer Given By Candidate:C

Question ID:100680
Topic Name:Chemistry-Section A
Question:

For kinetic study of the reaction of iodide ion with H_2O_2 at room temperature :

(A) Always use freshly prepared starch solution.

(B) Always keep the concentration of sodium thiosulphate solution less than that of KI solution.

(C) Record the time immediately after the appearance of blue colour.

(D) Record the time immediately before the appearance of blue colour.

(E) Always keep the concentration of sodium thiosulphate solution more than that of KI solution.

Choose the **correct** answer from the options given below :

A (A), (B), (C) only

B (A), (D), (E) only

C (D), (E) only

D (A), (B), (E) only

Answer Given By Candidate:A

Question ID:100681
Topic Name:Chemistry-Section B
Question:

In the given reaction,

$X + Y + 3Z \leftrightarrows XYZ_3$

if one mole of each of X and Y with 0.05 mol of Z gives compound XYZ_3. (Given : Atomic masses of X, Y and Z are 10, 20 and 30 amu, respectively.) The yield of XYZ_3 is _____g. (Nearest integer)

Answer Given By Candidate:2

Question ID:100682
Topic Name:Chemistry-Section B
Question:

An element M crystallises in a body centred cubic unit cell with a cell edge of 300 pm. The density of the element is 6.0 g cm^{-3}. The number of atoms present in 180 g of the element is _____ $\times 10^{23}$. (Nearest integer)

Answer Given By Candidate:Not

AttemptedQuestion ID:100683
Topic Name:Chemistry-Section B
Question:

The number of paramagnetic species among the following is _____.

$B_2, Li_2, C_2, C_2^-, O_2^{2-}, O_2^+$ and He_2^+

Answer Given By Candidate:3

Question ID:100684
Topic Name:Chemistry-Section B
Question:

150 g of acetic acid was contaminated with 10.2 g ascorbic acid ($C_6H_8O_6$) to lower down its freezing point by ($x \times 10^{-1}$)°C. The value of x is _____. (Nearest integer)

[Given K_f = 3.9 K kg mol^{-1};

molar mass of ascorbic acid = 176 g mol^{-1}]

Answer Given By Candidate:15

Question ID:100685
Topic Name:Chemistry-Section B

Question:

K_a for butyric acid (C_3H_7COOH) is 2×10^{-5}. The pH of 0.2 M solution of butyric acid is _____ $\times 10^{-1}$. (Nearest integer)

[Given log 2 = 0.30]

Answer Given By Candidate:54

Question ID:100686
Topic Name:Chemistry-Section B
Question:

For the given first order reaction

$A \rightarrow B$

the half life of the reaction is 0.3010 min. The ratio of the initial concentration of reactant to the concentration of reactant at time 2.0 min will be equal to _____. (Nearest integer)

Answer Given By Candidate:Not

AttemptedQuestion ID:100687
Topic Name:Chemistry-Section B
Question:

The number of interhalogens from the following having square pyramidal structure is :

ClF_3, IF_7, BrF_5, BrF_3, I_2Cl_6, IF_5, ClF, ClF_5

Answer Given By Candidate:3

Question ID:100688
Topic Name:Chemistry-Section B
Question:

The disproportionation of MnO_4^{2-} in acidic medium resulted in the formation of two manganese compounds A and B. If the oxidation state of Mn in B is smaller than that of A, then the spin-only magnetic moment (μ) value of B in BM is _____. (Nearest integer)

Answer Given By Candidate:Not

AttemptedQuestion ID:100689
Topic Name:Chemistry-Section B
Question:

Total number of relatively more stable isomer(s) possible for octahedral complex $[Cu(en)_2(SCN)_2]$ will be _____.

Answer Given By Candidate:Not

AttemptedQuestion ID:100690
Topic Name:Chemistry-Section B

Question:

On complete combustion of 0.492 g of an organic compound containing C, H and O, 0.7938 g of CO_2 and 0.4428 g of H_2O was produced. The % composition of oxygen in the compound is _____.

Answer Given By Candidate: Not Attempted